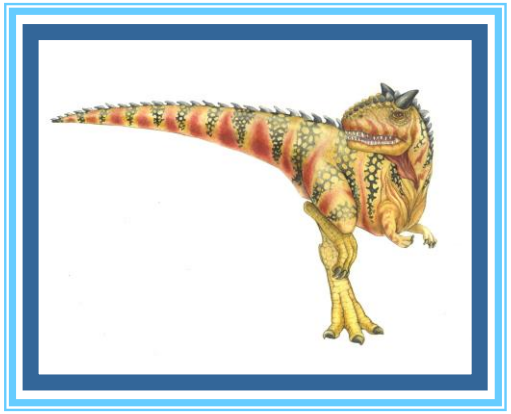
Chapter 9: Main Memory



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Chapter 9: Memory Management

Background

Swapping

Contiguous Memory Allocation

Segmentation

Paging

Structure of the Page Table

Example: The Intel 32 and 64-bit Architectures

Example: ARM Architecture

Objectives

To provide a detailed description of various ways of organizing memory hardware

To discuss various memory-management techniques, including paging and segmentation

To provide a detailed description of the Intel Pentium, which supports both pure segmentation and segmentation with paging

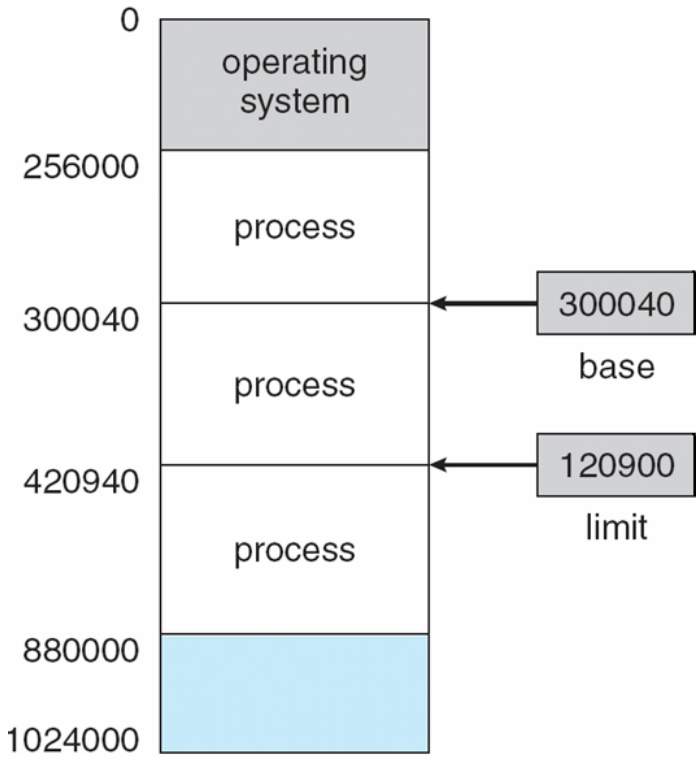
Background

- Program must be brought (from disk) into memory and placed within a process for it to be run
- Main memory and registers are only storage CPU can access directly
- Memory unit only sees a stream of addresses + read requests, or address + data and write requests
- Register access in one CPU clock (or less)
- Main memory can take many cycles, causing a **stall**
- **Cache** sits between main memory and CPU registers
- Protection of memory required to ensure correct operation

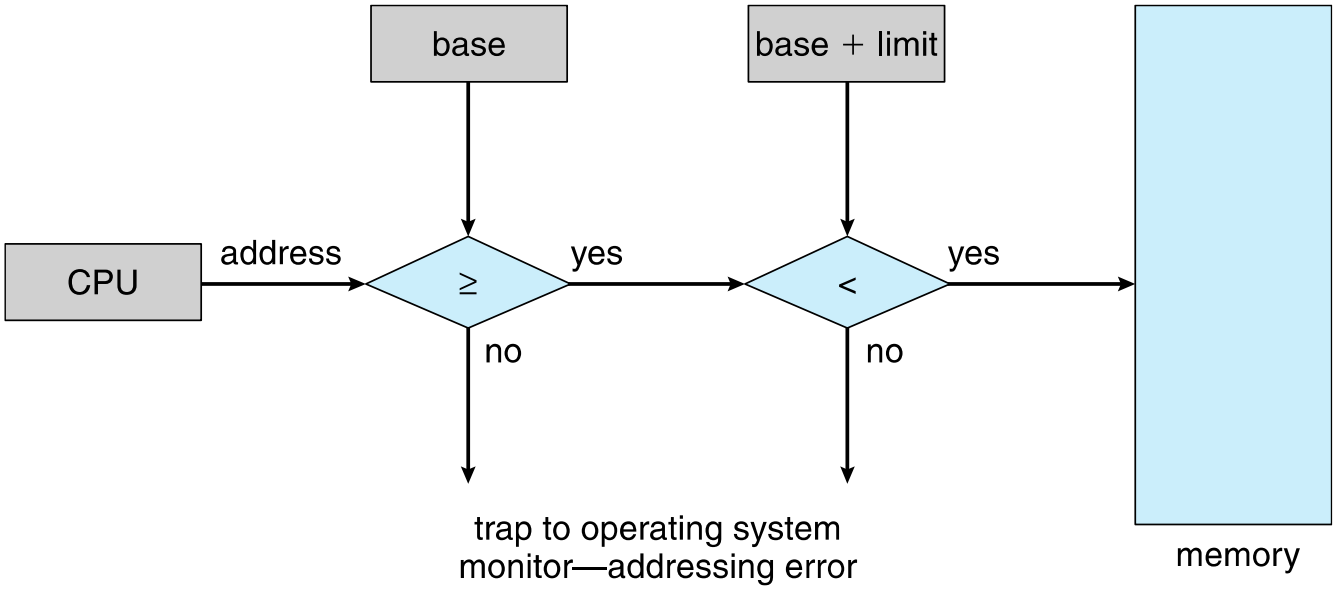
Base and Limit Registers

A pair of **base** and **limit registers** define the logical address space

CPU must check every memory access generated in user mode to be sure it is between base and limit for that user



Hardware Address Protection



Address Binding

Programs on disk, ready to be brought into memory to execute form an **input queue**

- Without support, must be loaded into address 0000

Inconvenient to have first user process physical address always at 0000

- How can it not be?

Further, addresses represented in different ways at different stages of a program's life

- Source code addresses usually symbolic
- Compiled code addresses **bind** to relocatable addresses
  - i.e. "14 bytes from beginning of this module"
- Linker or loader will bind relocatable addresses to absolute addresses
  - i.e. 74014
- Each binding maps one address space to another

Binding of Instructions and Data to Memory

Address binding of instructions and data to memory addresses can happen at three different stages

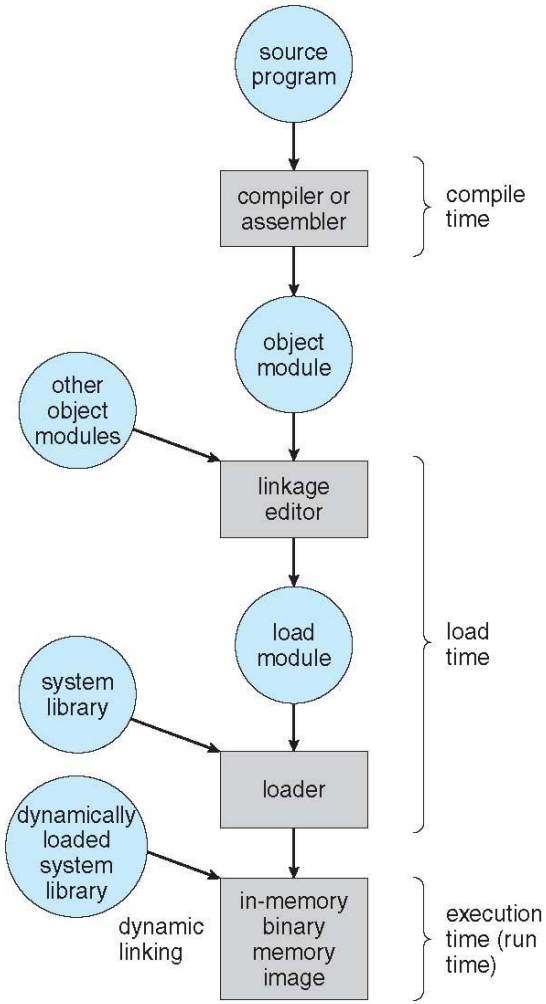
Compile time: If memory location known a priori, **absolute code** can be generated; must recompile code if starting location changes

Load time: Must generate **relocatable code** if memory location is not known at compile time

Execution time: Binding delayed until run time if the process can be moved during its execution from one memory segment to another

- Need hardware support for address maps (e.g., base and limit registers)

Multistep Processing of a User Program



Logical vs. Physical Address Space

The concept of a logical address space that is bound to a separate **physical address space** is central to proper memory management

- **Logical address** – generated by the CPU; also referred to as **virtual address**
- **Physical address** – address seen by the memory unit

Logical and physical addresses are the same in compile-time and load-time address-binding schemes; logical (virtual) and physical addresses differ in execution-time address-binding scheme

Logical address space is the set of all logical addresses generated by a program

Physical address space is the set of all physical addresses generated by a program

Memory-Management Unit (MMU)

Hardware device that at run time maps virtual to physical address

Many methods possible, covered in the rest of this chapter

To start, consider simple scheme where the value in the relocation register is added to every address generated by a user process at the time it is sent to memory

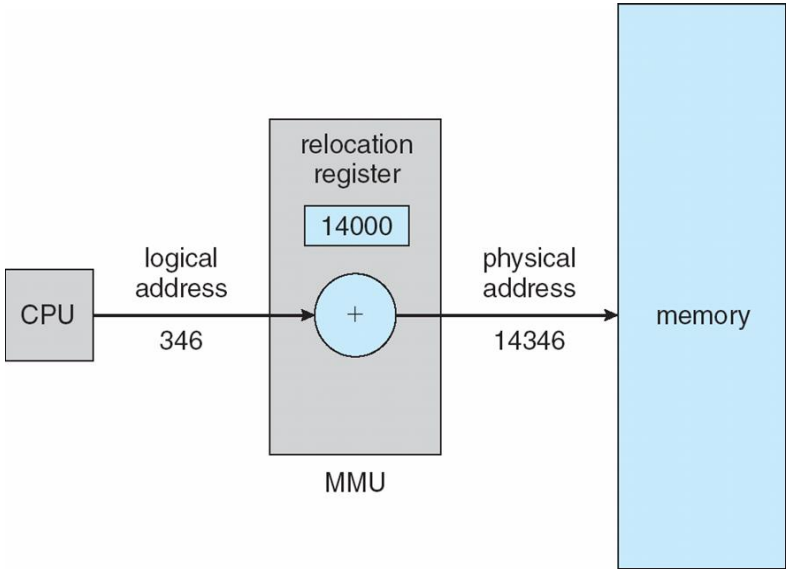
- Base register now called **relocation register**
- MS-DOS on Intel 80x86 used 4 relocation registers

The user program deals with *logical* addresses; it never sees the *real* physical addresses

- Execution-time binding occurs when reference is made to location in memory
- Logical address bound to physical addresses

Dynamic relocation using a relocation register

- Routine is not loaded until it is called
- Better memory-space utilization; unused routine is never loaded
- All routines kept on disk in relocatable load format
- Useful when large amounts of code are needed to handle infrequently occurring cases
- No special support from the operating system is required
  - Implemented through program design
  - OS can help by providing libraries to implement dynamic loading



Dynamic Linking

- **Static linking** – system libraries and program code combined by the loader into the binary program image
- Dynamic linking –linking postponed until execution time
- Small piece of code, **stub**, used to locate the appropriate memory-resident library routine
- Stub replaces itself with the address of the routine, and executes the routine
- Operating system checks if routine is in processes' memory address
  - If not in address space, add to address space
- Dynamic linking is particularly useful for libraries
- System also known as **shared libraries**
- Consider applicability to patching system libraries
  - Versioning may be needed

Swapping

A process can be **swapped** temporarily out of memory to a backing store, and then brought back into memory for continued execution

- Total physical memory space of processes can exceed physical memory

Backing store – fast disk large enough to accommodate copies of all memory images for all users; must provide direct access to these memory images

Roll out, roll in – swapping variant used for priority-based scheduling algorithms; lower-priority process is swapped out so higher-priority process can be loaded and executed

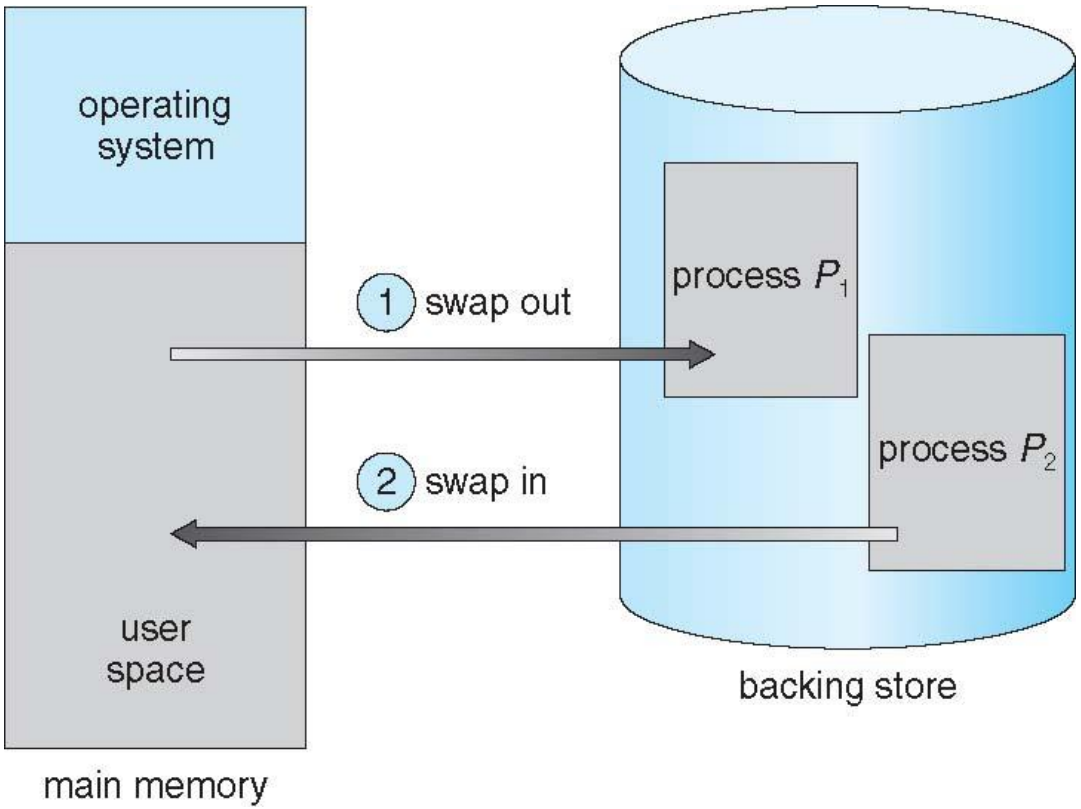
Major part of swap time is transfer time; total transfer time is directly proportional to the amount of memory swapped

System maintains a **ready queue** of ready-to-run processes which have memory images on disk

Swapping (Cont.)

- Does the swapped out process need to swap back in to same physical addresses?
- Depends on address binding method
  - Plus consider pending I/O to / from process memory space
- Modified versions of swapping are found on many systems (i.e., UNIX, Linux, and Windows)
  - Swapping normally disabled
  - Started if more than threshold amount of memory allocated
  - Disabled again once memory demand reduced below threshold

Schematic View of Swapping



Context Switch Time including Swapping

- If next processes to be put on CPU is not in memory, need to swap out a process and swap in target process
- Context switch time can then be very high
- 100MB process swapping to hard disk with transfer rate of 50MB/sec
  - Swap out time of 2000 ms
  - Plus swap in of same sized process
  - Total context switch swapping component time of 4000ms (4 seconds)
- Can reduce if reduce size of memory swapped – by knowing how much memory really being used
  - System calls to inform OS of memory use via `request_memory()` and `release_memory()`

Context Switch Time and Swapping (Cont.)

- Other constraints as well on swapping
  - Pending I/O – can't swap out as I/O would occur to wrong process
  - Or always transfer I/O to kernel space, then to I/O device
    - Known as **double buffering**, adds overhead
- Standard swapping not used in modern operating systems
  - But modified version common
    - Swap only when free memory extremely low

Swapping on Mobile Systems

- Not typically supported
  - Flash memory based
    - Small amount of space
    - Limited number of write cycles
    - Poor throughput between flash memory and CPU on mobile platform
- Instead use other methods to free memory if low
  - iOS ***asks*** apps to voluntarily relinquish allocated memory
    - Read-only data thrown out and reloaded from flash if needed
    - Failure to free can result in termination
  - Android terminates apps if low free memory, but first writes **application state** to flash for fast restart
  - Both OSes support paging as discussed below

Contiguous Allocation

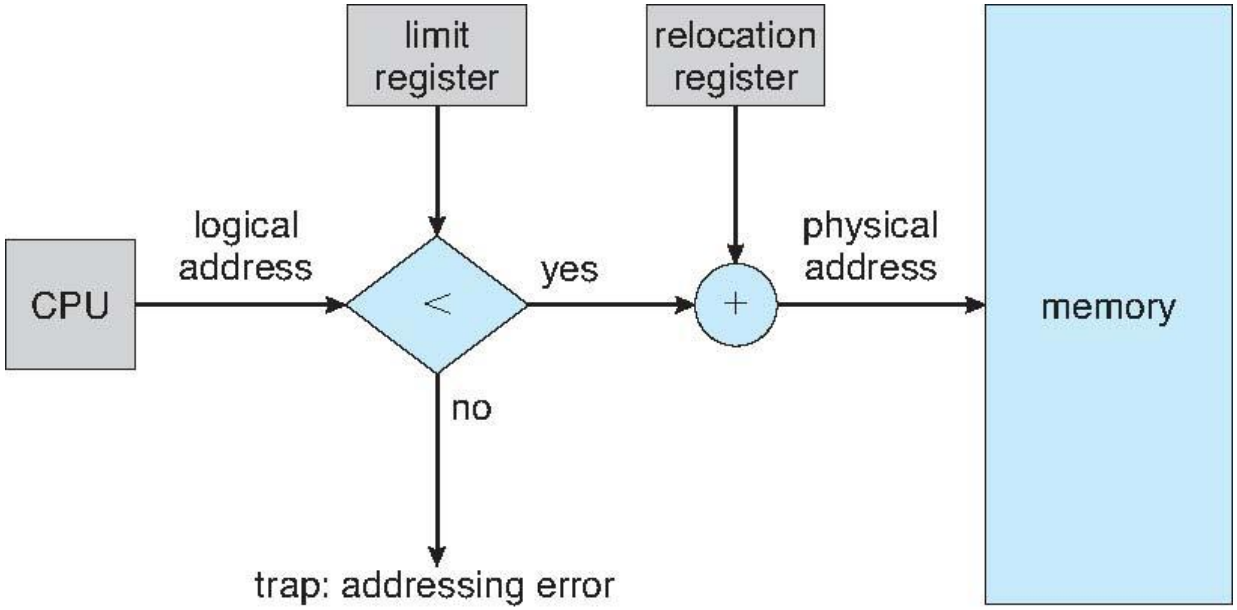
- Main memory must support both OS and user processes
- Limited resource, must allocate efficiently
- Contiguous allocation is one early method
- Main memory usually into two **partitions**:
  - Resident operating system, usually held in low memory with interrupt vector
  - User processes then held in high memory
  - Each process contained in single contiguous section of memory

Contiguous Allocation (Cont.)

Relocation registers used to protect user processes from each other, and from changing operating-system code and data

- Base register contains value of smallest physical address
- Limit register contains range of logical addresses – each logical address must be less than the limit register
- MMU maps logical address *dynamically*
- Can then allow actions such as kernel code being **transient** and kernel changing size

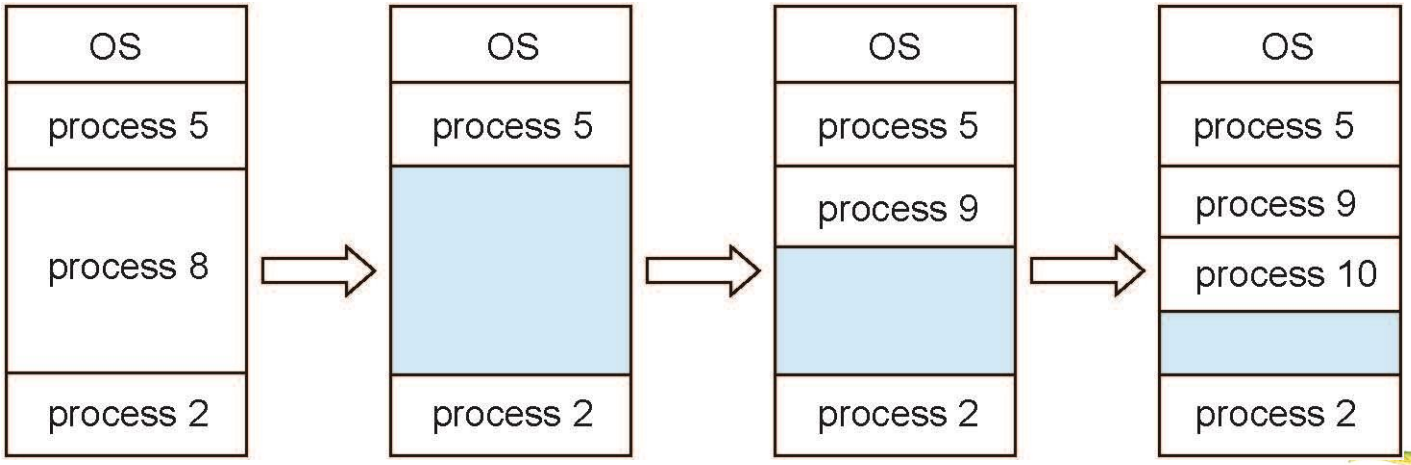
Hardware Support for Relocation and Limit Registers



Multiple-partition allocation

Multiple-partition allocation

- Degree of multiprogramming limited by number of partitions
- **Variable-partition** sizes for efficiency (sized to a given process' needs)
- **Hole** – block of available memory; holes of various size are scattered throughout memory
- When a process arrives, it is allocated memory from a hole large enough to accommodate it
- Process exiting frees its partition, adjacent free partitions combined
- Operating system maintains information about:
  a) allocated partitions b) free partitions (hole)



Dynamic Storage-Allocation Problem

How to satisfy a request of size ***n*** from a list of free holes?

First-fit: Allocate the ***first*** hole that is big enough

Best-fit: Allocate the ***smallest*** hole that is big enough; must search entire list, unless ordered by size

Produces the smallest leftover hole

Worst-fit: Allocate the ***largest*** hole; must also search entire list

Produces the largest leftover hole

First-fit and best-fit better than worst-fit in terms of speed and storage utilization

Fragmentation

External Fragmentation – total memory space exists to satisfy a request, but it is not contiguous

Internal Fragmentation – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used

First fit analysis reveals that given *N* blocks allocated, 0.5 *N* blocks lost to fragmentation

1/3 may be unusable -> **50-percent rule**

Fragmentation (Cont.)

Reduce external fragmentation by **compaction**

Shuffle memory contents to place all free memory together in one large block

Compaction is possible *only* if relocation is dynamic, and is done at execution time

I/O problem

- Latch job in memory while it is involved in I/O
- Do I/O only into OS buffers

Now consider that backing store has same fragmentation problems

Segmentation

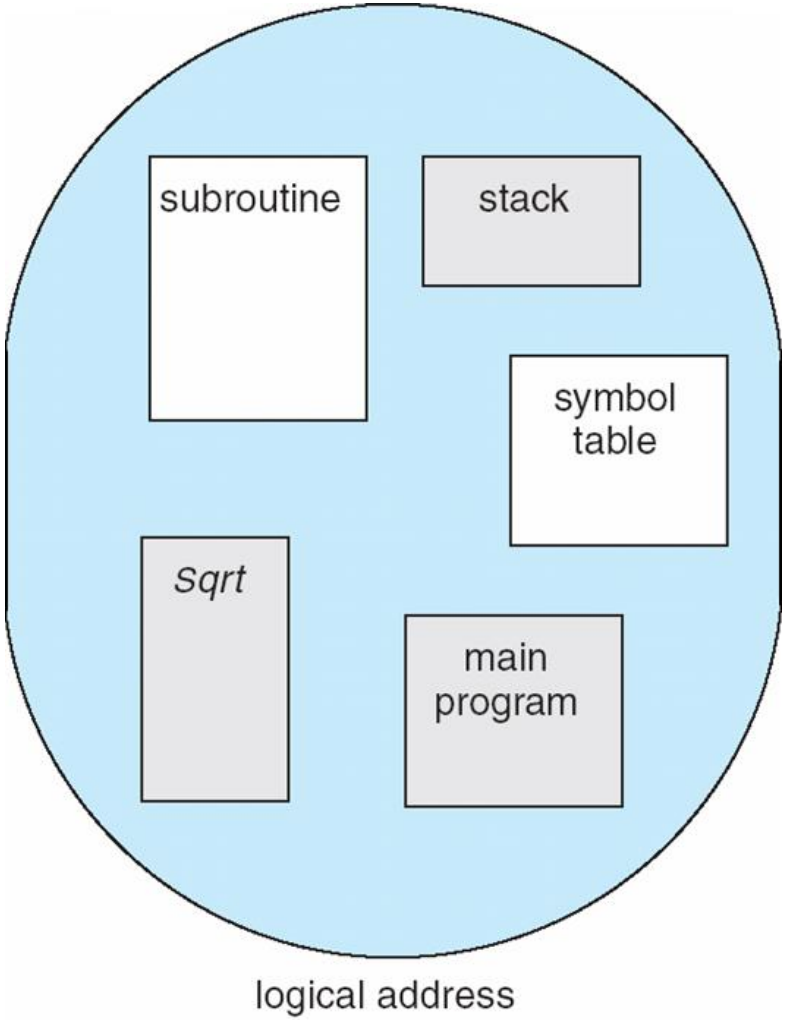
Memory-management scheme that supports user view of memory

A program is a collection of segments

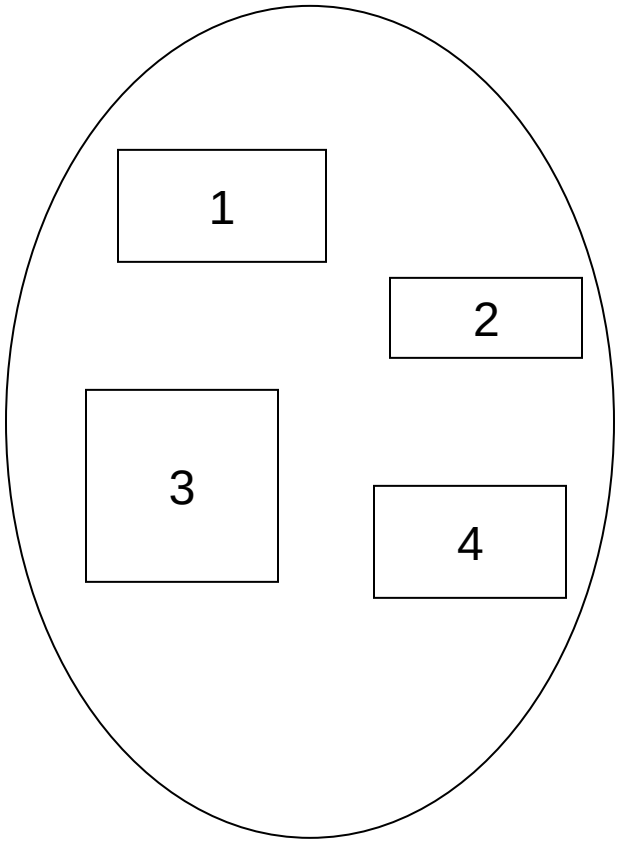
A segment is a logical unit such as:

- main program
- procedure
- function
- method
- object
- local variables, global variables
- common block
- stack
- symbol table
- arrays

User's View of a Program



Logical View of Segmentation



user space

physical memory space

Segmentation Architecture

Logical address consists of a two tuple:

<segment-number, offset>,

Segment table – maps two-dimensional physical addresses; each table entry has:

- **base** – contains the starting physical address where the segments reside in memory
- **limit** – specifies the length of the segment

Segment-table base register (STBR) points to the segment table's location in memory

Segment-table length register (STLR) indicates number of segments used by a program;

segment number ***s*** is legal if ***s*** < **STLR**

Segmentation Architecture (Cont.)

Protection

With each entry in segment table associate:

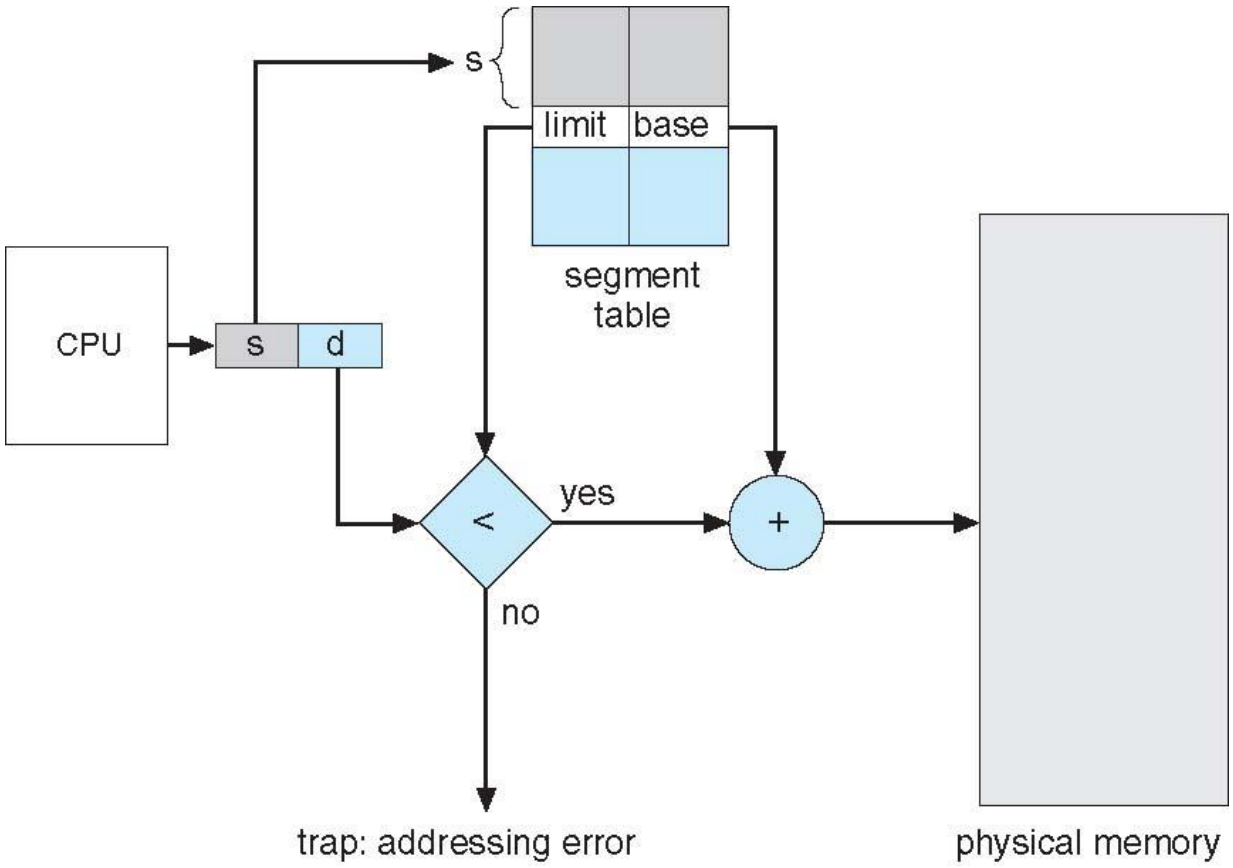
- validation bit = 0 $\Rightarrow$ illegal segment
- read/write/execute privileges

Protection bits associated with segments; code sharing occurs at segment level

Since segments vary in length, memory allocation is a dynamic storage-allocation problem

A segmentation example is shown in the following diagram

Segmentation Hardware



Paging

Physical address space of a process can be noncontiguous; process is allocated physical memory whenever the latter is available

- Avoids external fragmentation
- Avoids problem of varying sized memory chunks

Divide physical memory into fixed-sized blocks called **frames**

- Size is power of 2, between 512 bytes and 16 Mbytes

Divide logical memory into blocks of same size called **pages**

Keep track of all free frames

To run a program of size ***N*** pages, need to find ***N*** free frames and load program

Set up a **page table** to translate logical to physical addresses

Backing store likewise split into pages

Still have Internal fragmentation

Address Translation Scheme

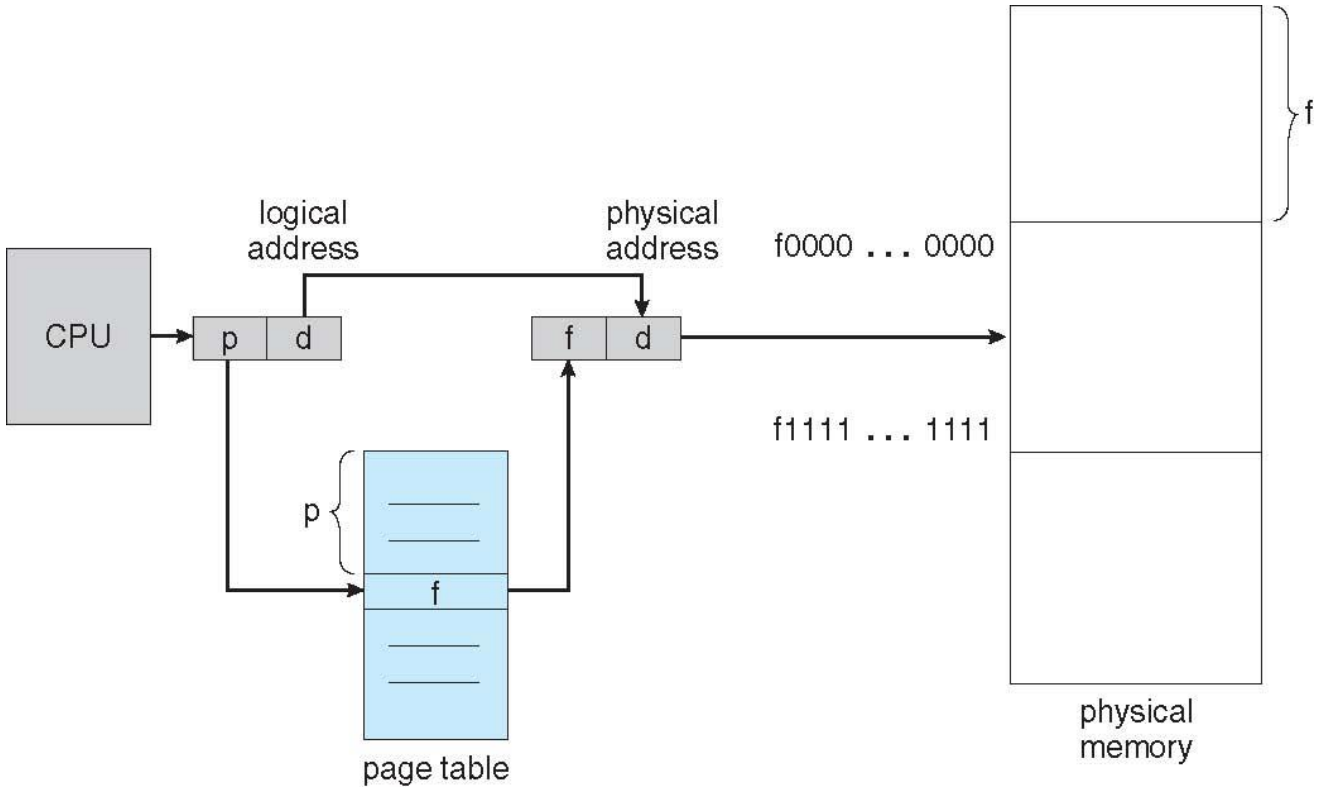
Address generated by CPU is divided into:

- **Page number** (***p***) – used as an index into a **page table** which contains base address of each page in physical memory
- **Page offset** (***d***) – combined with base address to define the physical memory address that is sent to the memory unit

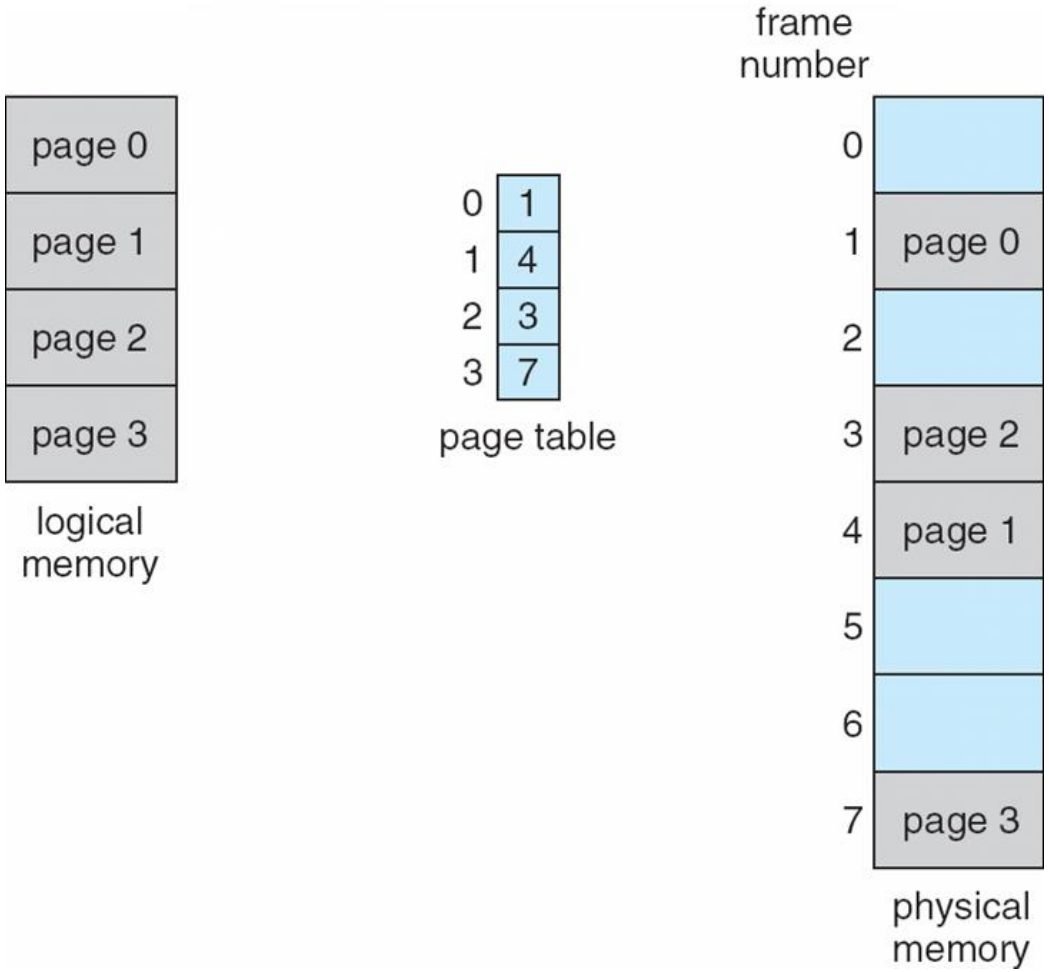
page number	page offset
p	d
m -n	n

For given logical address space 2^m and page size 2^n

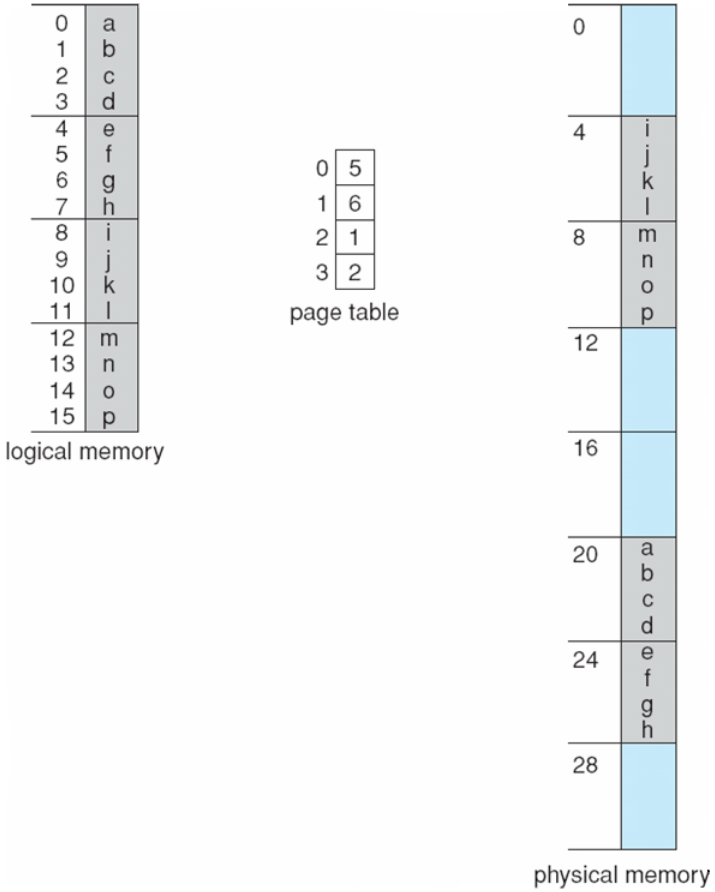
Paging Hardware



Paging Model of Logical and Physical Memory



Paging Example



n=2 and *m*=4 32-byte memory and 4-byte pages

Paging (Cont.)

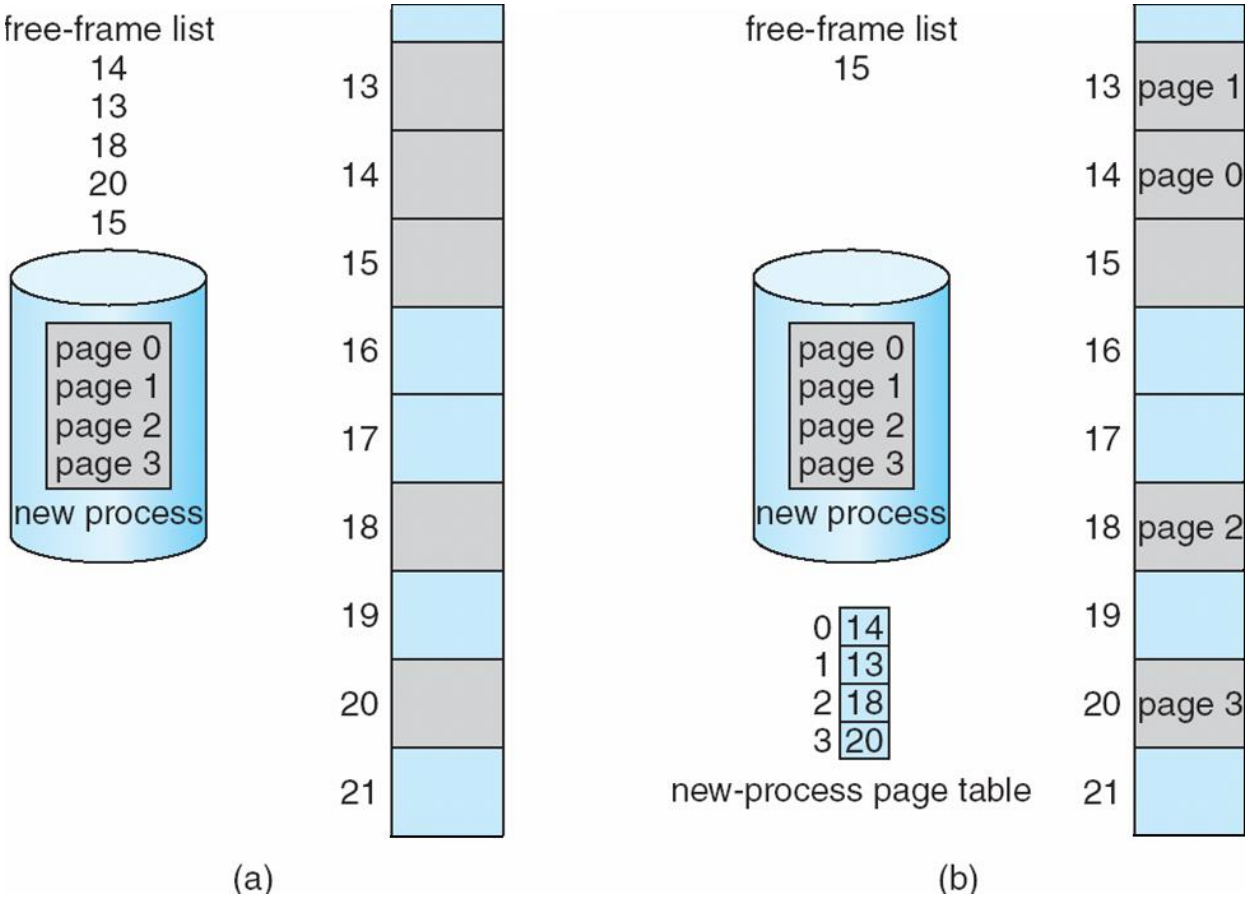
Calculating internal fragmentation

- Page size = 2,048 bytes
- Process size = 72,766 bytes
- 35 pages + 1,086 bytes
- Internal fragmentation of 2,048 - 1,086 = 962 bytes
- Worst case fragmentation = 1 frame – 1 byte
- On average fragmentation = 1 / 2 frame size
- So small frame sizes desirable?
- But each page table entry takes memory to track
- Page sizes growing over time
  - Solaris supports two page sizes – 8 KB and 4 MB

Process view and physical memory now very different

By implementation process can only access its own memory

Free Frames



Before allocation

After allocation

Implementation of Page Table

Page table is kept in main memory

Page-table base register (**PTBR**) points to the page table

Page-table length register (**PTLR**) indicates size of the page table

In this scheme every data/instruction access requires two memory accesses

One for the page table and one for the data / instruction

The two memory access problem can be solved by the use of a special fast-lookup hardware cache called **associative memory** or **translation look-aside buffers** (**TLBs**)

Implementation of Page Table (Cont.)

Some TLBs store **address-space identifiers** (**ASIDs**) in each TLB entry – uniquely identifies each process to provide address-space protection for that process

- Otherwise need to flush at every context switch

TLBs typically small (64 to 1,024 entries)

On a TLB miss, value is loaded into the TLB for faster access next time

- Replacement policies must be considered
- Some entries can be **wired down** for permanent fast access

Associative Memory

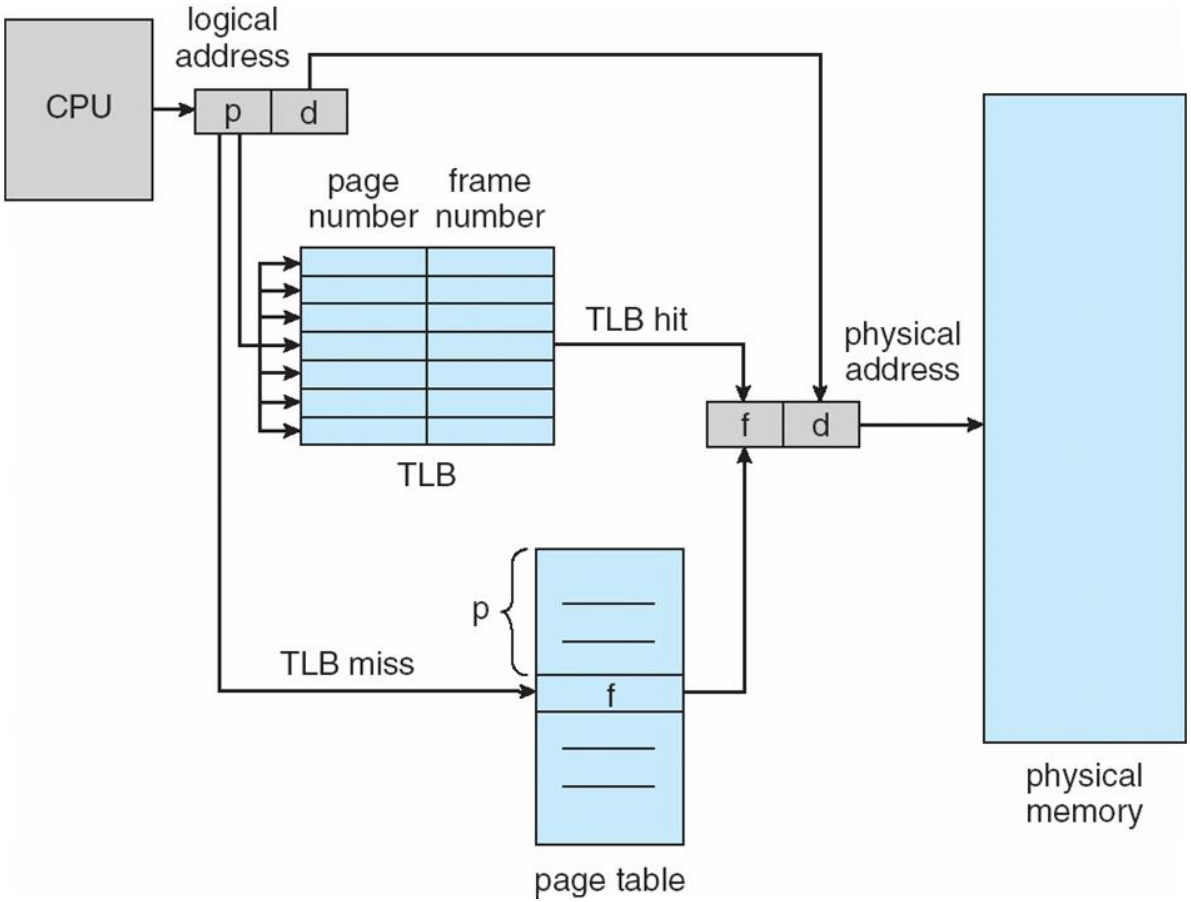
Associative memory – parallel search

Page #	Frame #

Address translation (p, d)

- If p is in associative register, get frame # out
- Otherwise get frame # from page table in memory

Paging Hardware With TLB



Effective Access Time

Associative Lookup = ε time unit

Can be < 10% of memory access time

Hit ratio = α

Hit ratio – percentage of times that a page number is found in the associative registers; ratio related to number of associative registers

Consider α = 80%, ε = 20ns for TLB search, 100ns for memory access

Effective Access Time (**EAT**)

$$\text{EAT} = (1 + \varepsilon)\,\alpha + (2 + \varepsilon)(1 - \alpha)$$
$$= 2 + \varepsilon - \alpha$$

Consider α = 80%, ε = 20ns for TLB search, 100ns for memory access

EAT = 0.80 x 100 + 0.20 x 200 = 120ns

Consider more realistic hit ratio -> α = 99%, ε = 20ns for TLB search, 100ns for memory access

EAT = 0.99 x 100 + 0.01 x 200 = 101ns

Memory Protection

Memory protection implemented by associating protection bit with each frame to indicate if read-only or read-write access is allowed

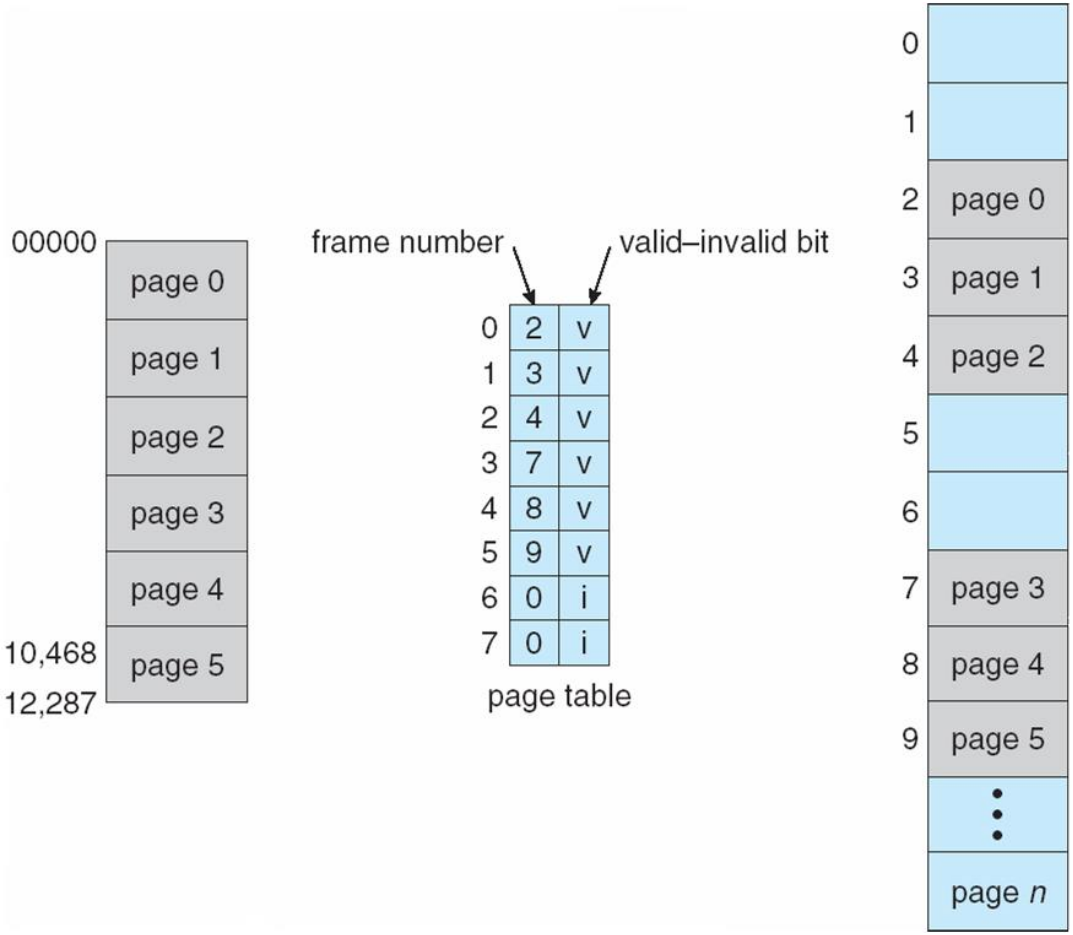
- Can also add more bits to indicate page execute-only, and so on

Valid-invalid bit attached to each entry in the page table:

- "valid" indicates that the associated page is in the process' logical address space, and is thus a legal page
- "invalid" indicates that the page is not in the process' logical address space
- Or use **page-table length register** (**PTLR**)

Any violations result in a trap to the kernel

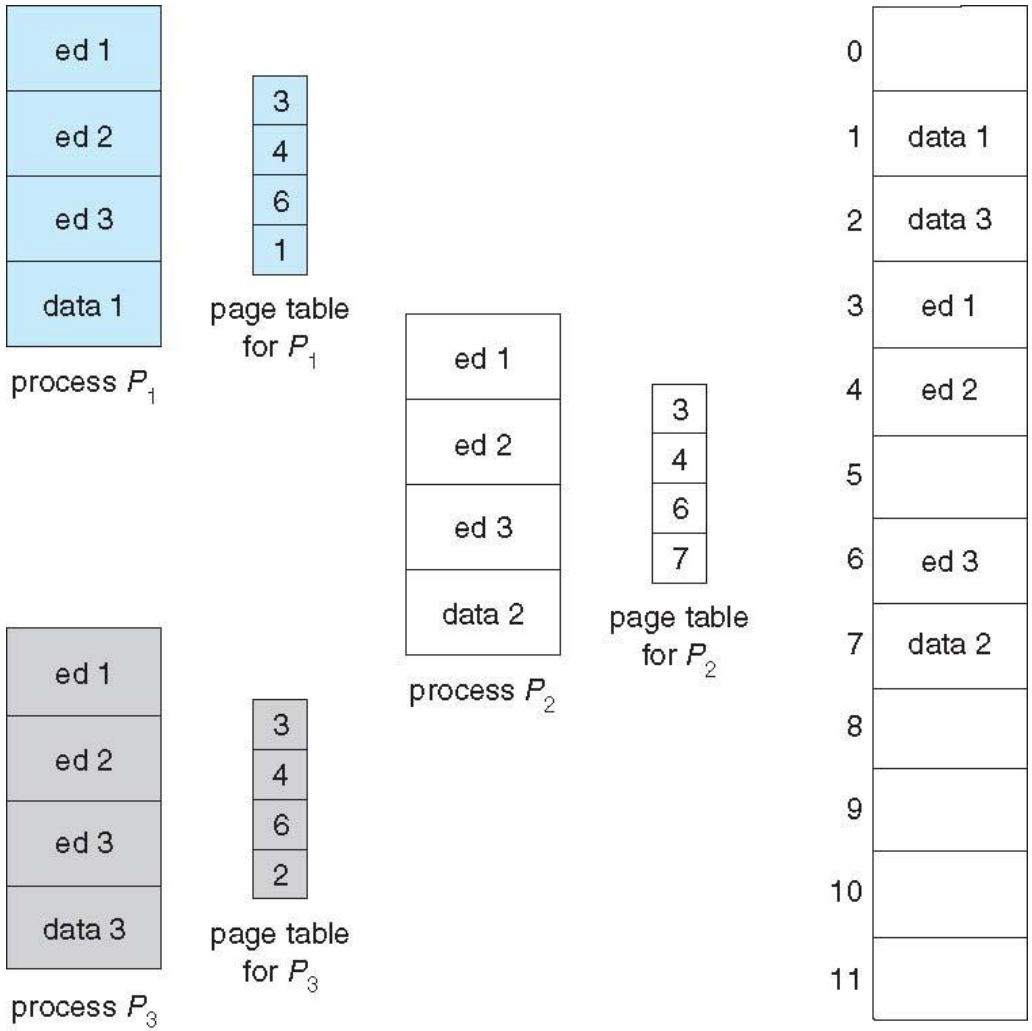
Valid (v) or Invalid (i) Bit In A Page Table



Shared Pages

- **Shared code**
  - One copy of read-only (**reentrant**) code shared among processes (i.e., text editors, compilers, window systems)
  - Similar to multiple threads sharing the same process space
  - Also useful for interprocess communication if sharing of read-write pages is allowed
- **Private code and data**
  - Each process keeps a separate copy of the code and data
  - The pages for the private code and data can appear anywhere in the logical address space

Shared Pages Example



Structure of the Page Table

Memory structures for paging can get huge using straight-forward methods

- Consider a 32-bit logical address space as on modern computers
- Page size of 4 KB (2^{12})
- Page table would have 1 million entries (2^{32} / 2^{12})
- If each entry is 4 bytes -> 4 MB of physical address space / memory for page table alone
  - That amount of memory used to cost a lot
  - Don't want to allocate that contiguously in main memory

Hierarchical Paging

Hashed Page Tables

Inverted Page Tables

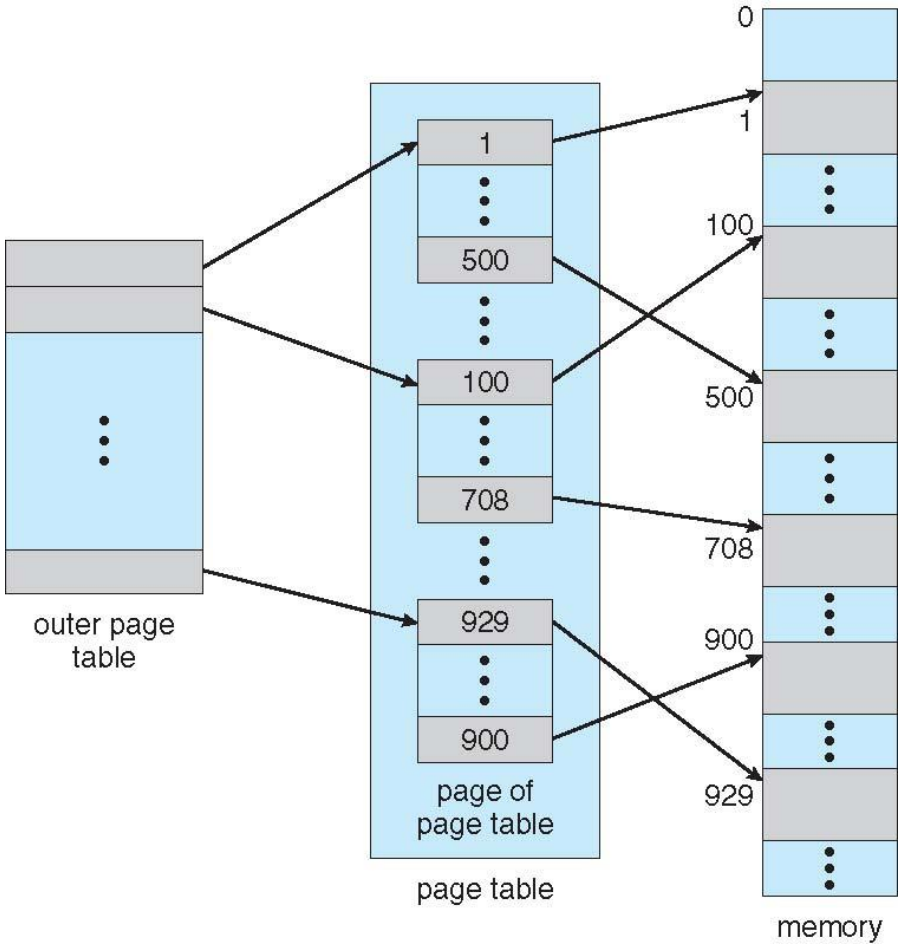
Hierarchical Page Tables

Break up the logical address space into multiple page tables

A simple technique is a two-level page table

We then page the page table

Two-Level Page-Table Scheme



Two-Level Paging Example

A logical address (on 32-bit machine with 1K page size) is divided into:

- a page number consisting of 22 bits
- a page offset consisting of 10 bits

Since the page table is paged, the page number is further divided into:

- a 12-bit page number
- a 10-bit page offset

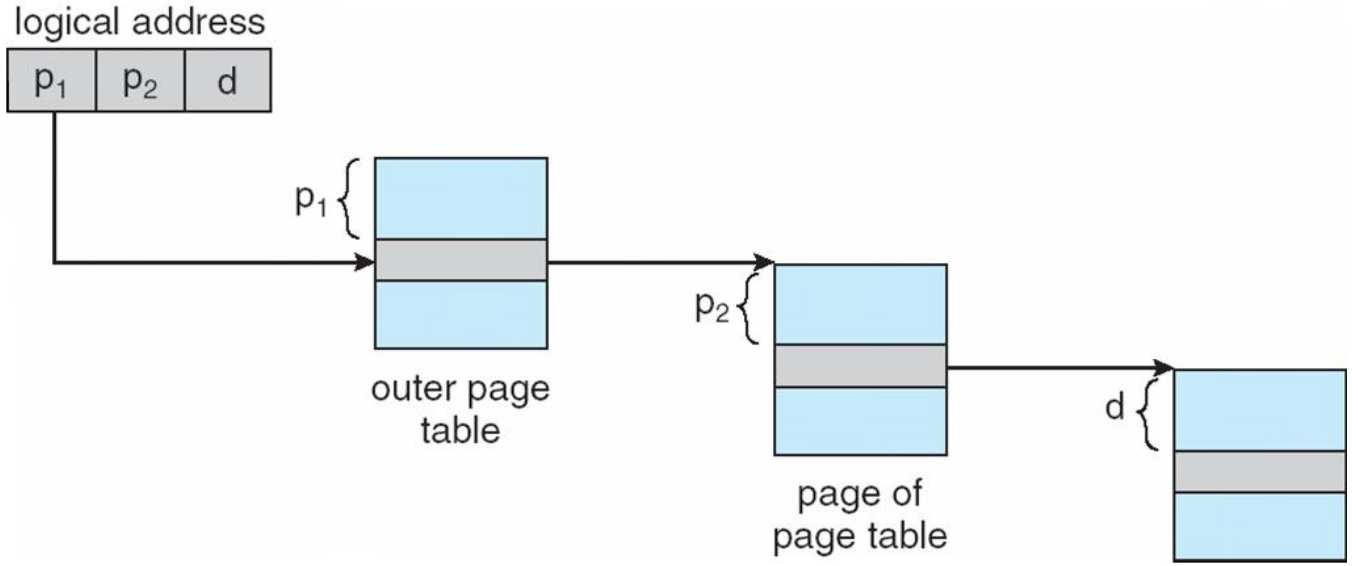
Thus, a logical address is as follows:

page number		page offset
p_1	p_2	d
12	10	10

where p_1 is an index into the outer page table, and p_2 is the displacement within the page of the inner page table

Known as **forward-mapped page table**

Address-Translation Scheme



64-bit Logical Address Space

Even two-level paging scheme not sufficient

If page size is 4 KB (2^{12})

Then page table has 2^{52} entries

If two level scheme, inner page tables could be 2^{10} 4-byte entries

Address would look like

outer page	inner page	page offset
p_1	p_2	d
42	10	12

Outer page table has 2^{42} entries or 2^{44} bytes

One solution is to add a 2nd outer page table

But in the following example the 2nd outer page table is still 2^{34} bytes in size

- And possibly 4 memory access to get to one physical memory location

Three-level Paging Scheme

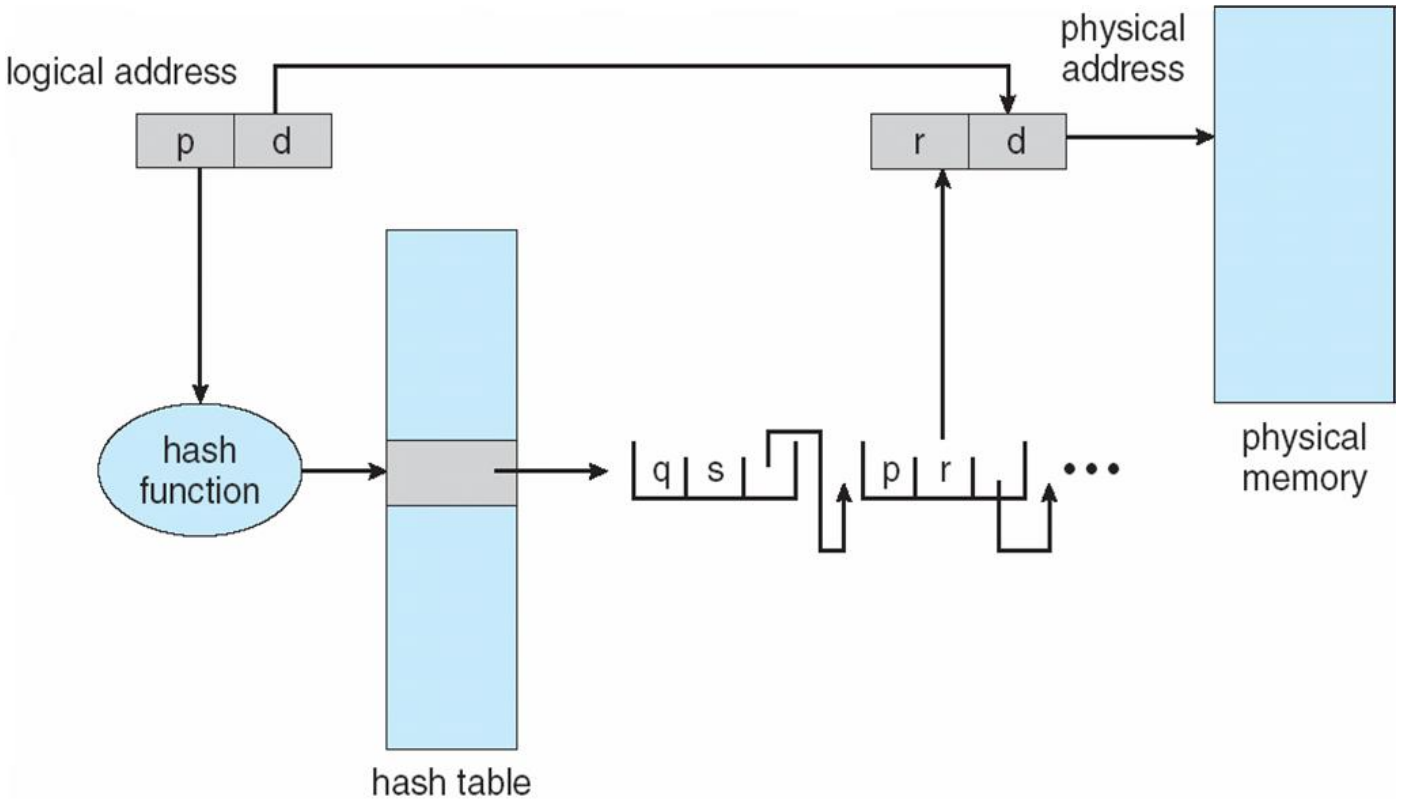
outer page	inner page	offset
p_1	p_2	d
42	10	12

2nd outer page	outer page	inner page	offset
p_1	p_2	p_3	d
32	10	10	12

Hashed Page Tables

- Common in address spaces > 32 bits
- The virtual page number is hashed into a page table
  - This page table contains a chain of elements hashing to the same location
- Each element contains (1) the virtual page number (2) the value of the mapped page frame (3) a pointer to the next element
- Virtual page numbers are compared in this chain searching for a match
  - If a match is found, the corresponding physical frame is extracted
- Variation for 64-bit addresses is **clustered page tables**
  - Similar to hashed but each entry refers to several pages (such as 16) rather than 1
  - Especially useful for **sparse** address spaces (where memory references are non-contiguous and scattered)

Hashed Page Table



Inverted Page Table

Rather than each process having a page table and keeping track of all possible logical pages, track all physical pages

One entry for each real page of memory

Entry consists of the virtual address of the page stored in that real memory location, with information about the process that owns that page

Decreases memory needed to store each page table, but increases time needed to search the table when a page reference occurs

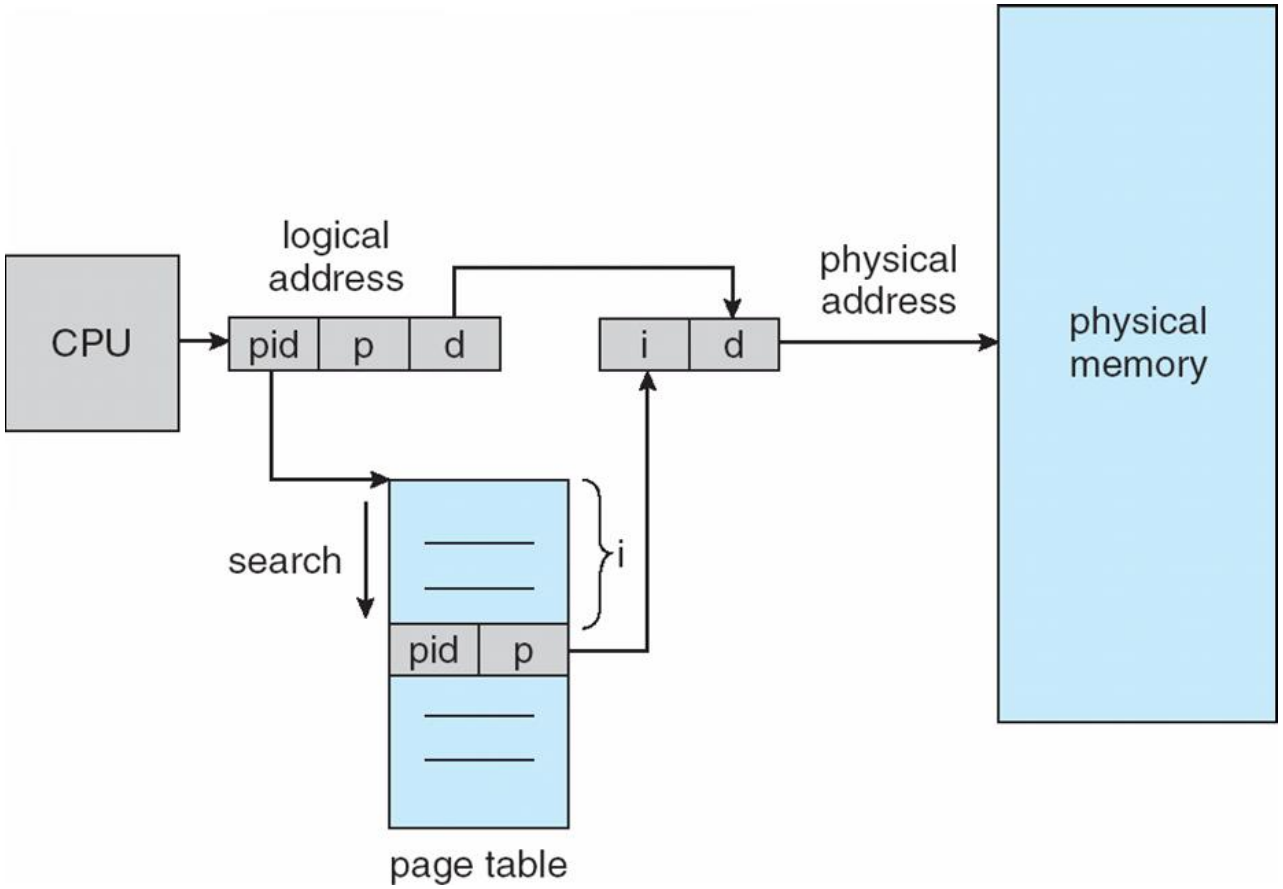
Use hash table to limit the search to one — or at most a few — page-table entries

- TLB can accelerate access

But how to implement shared memory?

- One mapping of a virtual address to the shared physical address

Inverted Page Table Architecture



Oracle SPARC Solaris

- Consider modern, 64-bit operating system example with tightly integrated HW
  - Goals are efficiency, low overhead
- Based on hashing, but more complex
- Two hash tables
  - One kernel and one for all user processes
  - Each maps memory addresses from virtual to physical memory
  - Each entry represents a contiguous area of mapped virtual memory,
    - More efficient than having a separate hash-table entry for each page
  - Each entry has base address and span (indicating the number of pages the entry represents)

Oracle SPARC Solaris (Cont.)

- TLB holds translation table entries (TTEs) for fast hardware lookups
  - A cache of TTEs reside in a translation storage buffer (TSB)
    - Includes an entry per recently accessed page
- Virtual address reference causes TLB search
  - If miss, hardware walks the in-memory TSB looking for the TTE corresponding to the address
    - If match found, the CPU copies the TSB entry into the TLB and translation completes
    - If no match found, kernel interrupted to search the hash table
      - The kernel then creates a TTE from the appropriate hash table and stores it in the TSB, Interrupt handler returns control to the MMU, which completes the address translation.

Example: The Intel 32 and 64-bit Architectures

Dominant industry chips

Pentium CPUs are 32-bit and called IA-32 architecture

Current Intel CPUs are 64-bit and called IA-64 architecture

Many variations in the chips, cover the main ideas here

Example: The Intel IA-32 Architecture

Supports both segmentation and segmentation with paging

- Each segment can be 4 GB
- Up to 16 K segments per process
- Divided into two partitions
  - First partition of up to 8 K segments are private to process (kept in **local descriptor table** (**LDT**))
  - Second partition of up to 8K segments shared among all processes (kept in **global descriptor table** (**GDT**))

Example: The Intel IA-32 Architecture (Cont.)

CPU generates logical address

Selector given to segmentation unit

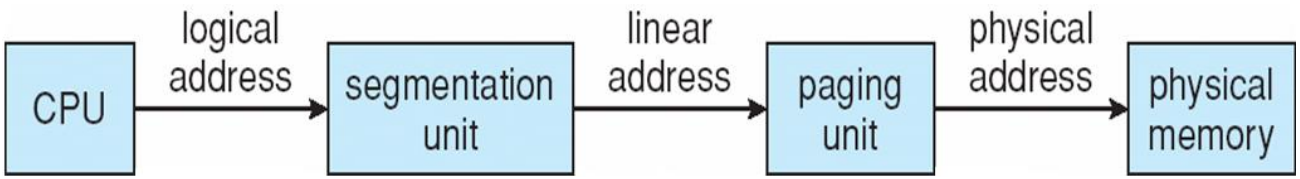
- Which produces linear addresses

s	g	p
13	1	2

Linear address given to paging unit

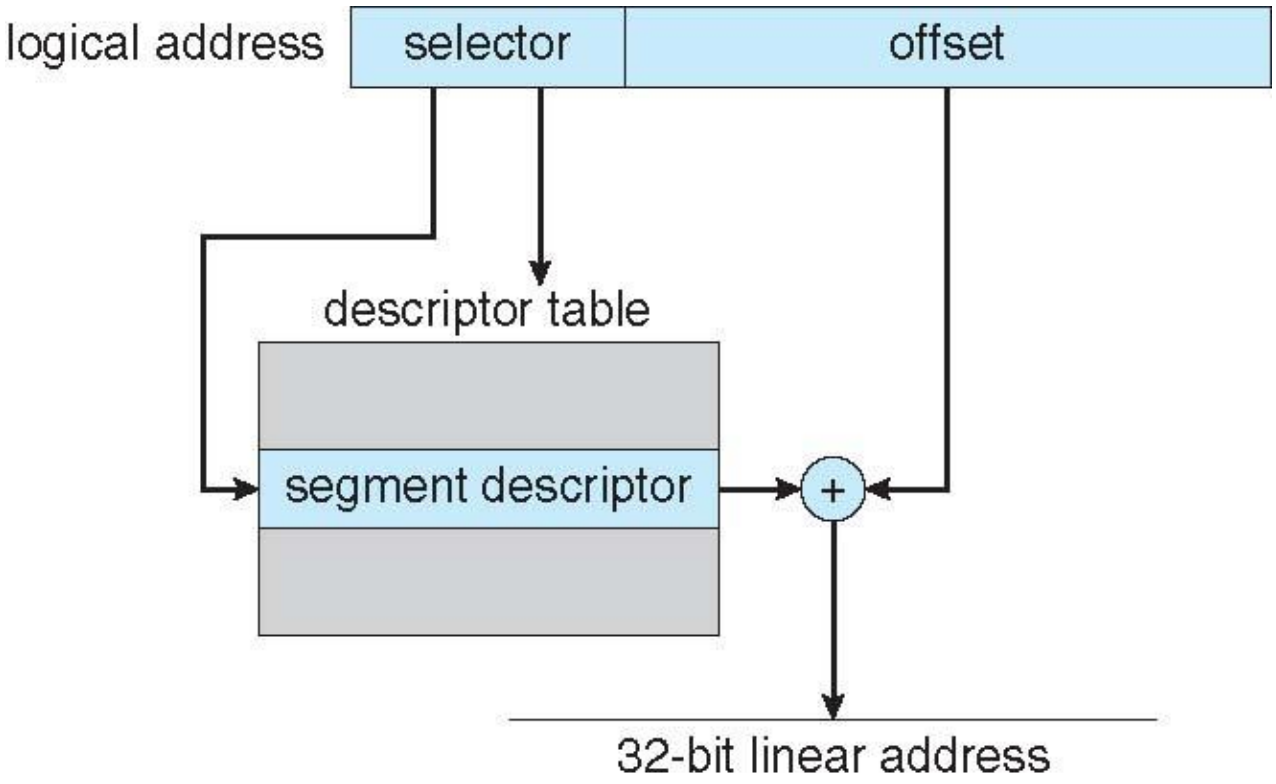
- Which generates physical address in main memory
- Paging units form equivalent of MMU
- Pages sizes can be 4 KB or 4 MB

Logical to Physical Address Translation in IA-32

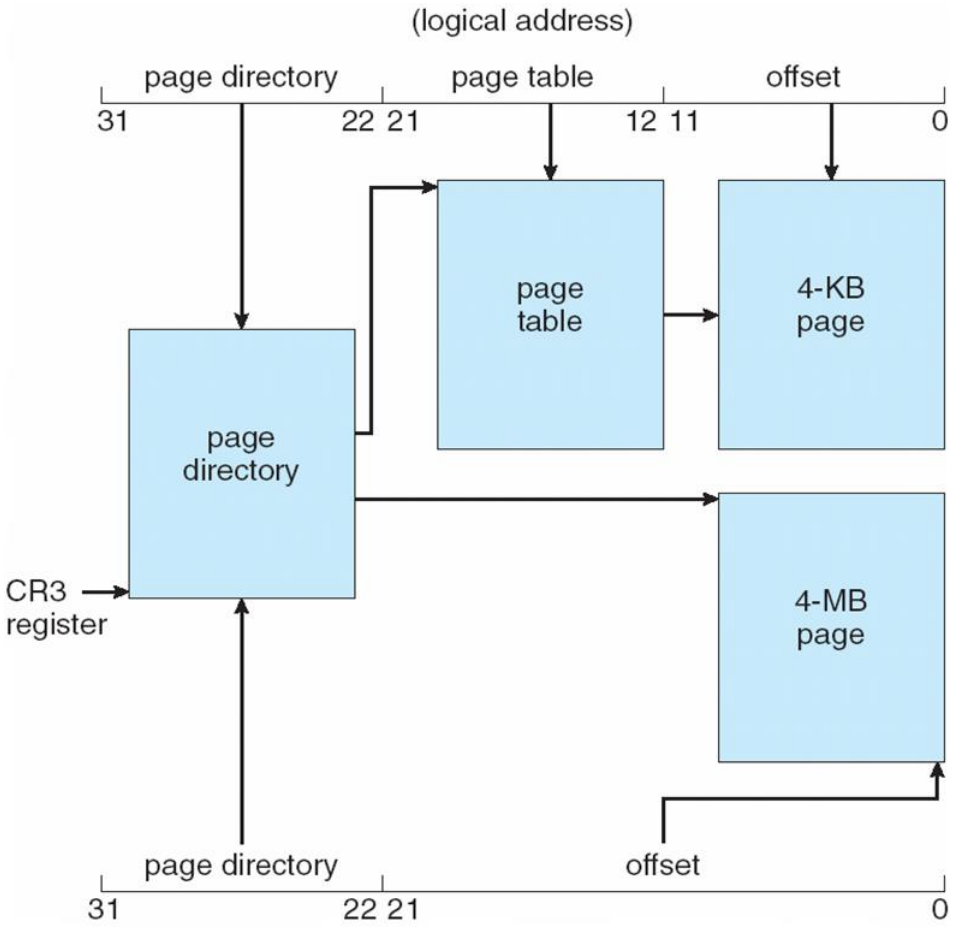


page number		page offset
p_1	p_2	d
10	10	12

Intel IA-32 Segmentation



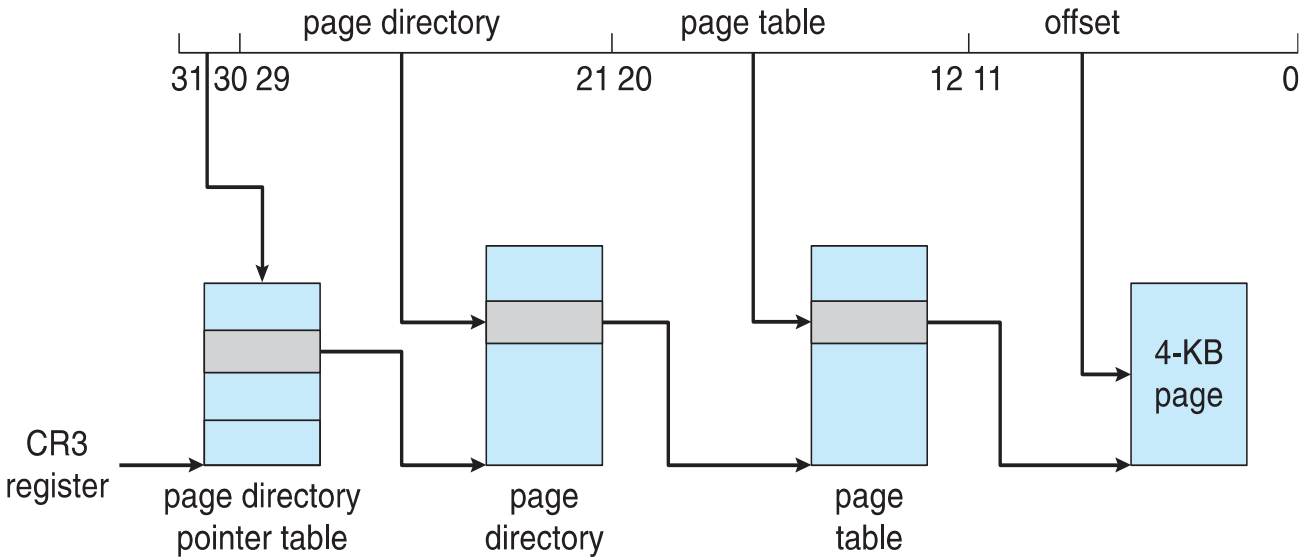
Intel IA-32 Paging Architecture



Intel IA-32 Page Address Extensions

32-bit address limits led Intel to create **page address extension** (**PAE**), allowing 32-bit apps access to more than 4GB of memory space

- Paging went to a 3-level scheme
- Top two bits refer to a **page directory pointer table**
- Page-directory and page-table entries moved to 64-bits in size
- Net effect is increasing address space to 36 bits – 64GB of physical memory



Intel x86-64

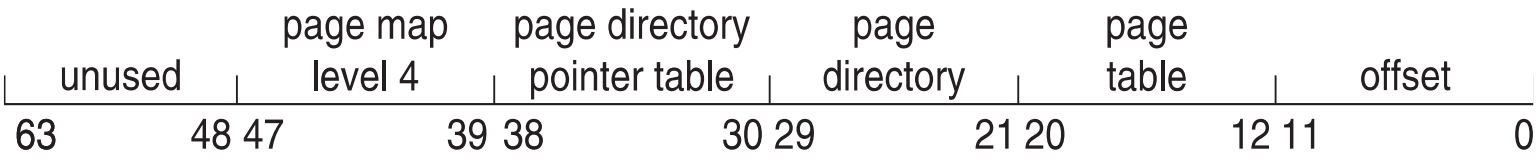
Current generation Intel x86 architecture

64 bits is ginormous (> 16 exabytes)

In practice only implement 48 bit addressing

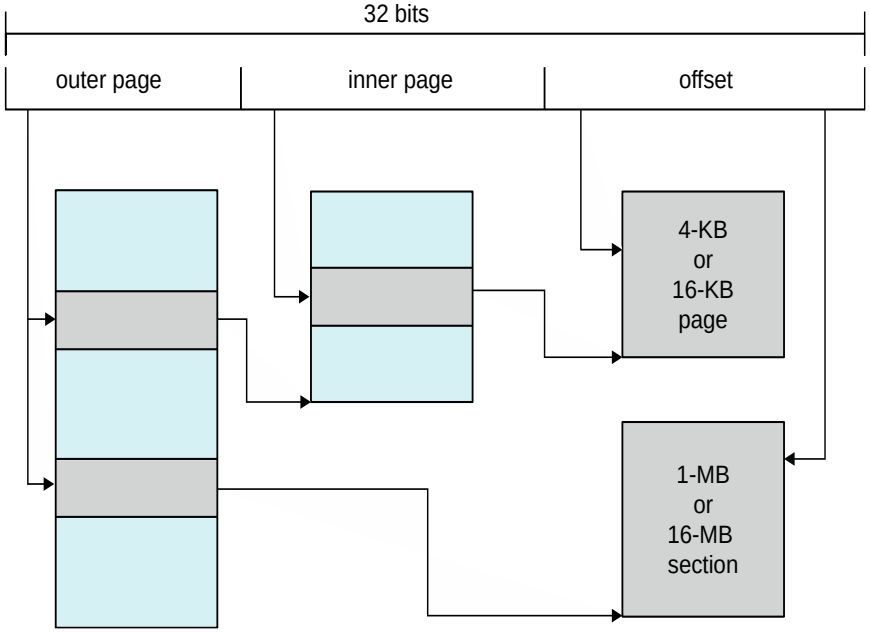
- Page sizes of 4 KB, 2 MB, 1 GB
- Four levels of paging hierarchy

Can also use PAE so virtual addresses are 48 bits and physical addresses are 52 bits

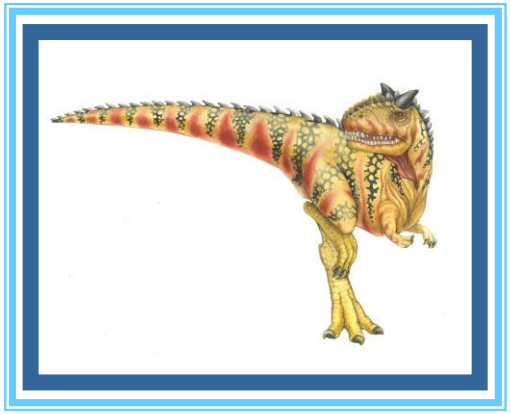


Example: ARM Architecture

- Dominant mobile platform chip (Apple iOS and Google Android devices for example)
- Modern, energy efficient, 32-bit CPU
- 4 KB and 16 KB pages
- 1 MB and 16 MB pages (termed **sections**)
- One-level paging for sections, two-level for smaller pages
- Two levels of TLBs
  - Outer level has two micro TLBs (one data, one instruction)
  - Inner is single main TLB
  - First inner is checked, on miss outers are checked, and on miss page table walk performed by CPU



End of Chapter 9



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